

I'm a results-driven Software Engineer with expertise in AI/ML, AR/VR, Cloud Computing, and Mobile Application Development. I have a proven ability to design, develop, and deploy scalable, high-performance solutions enhancing user experience and improving operational efficiency, as well as a strong foundation in Computer Science: with advanced skills in Python, C++, C#, SQL, Flutter, and cloud platforms. I'm also adept at applying research to real-world applications, driving innovation, and collaborating with cross-functional teams in fast-paced environments. I'm especially passionate about building immersive applications with a global reach: especially in healthcare, education, environment, gaming, and arts.

Work Experience

AI Engineer Intern AI Team	Clear Current USA (Remote)	Jun 2025 - Sep 2025
- Developed and automated end-to-end Python/MongoDB data pipelines for large-scale energy invoice validation, driving measurable improvements in operational efficiency and accuracy. - Evaluated, benchmarked, and fine-tuned multiple large language models (LLMs), finding the most cost-effective architectures for agentic AI workflows, and improving overall system performance, scalability, and cost efficiency. - Synthesized datasets using advanced Python libraries to simulate realistic energy consumption invoices, improving the robustness of training, validation, and testing pipelines. - Collaborated with cross-functional teams to integrate AI-driven solutions into enterprise platforms: seamless deployment and alignment with strategic business objectives. - Leveraged cloud-based platforms to develop, deploy, and scale AI/ML solutions, reinforcing system reliability, adherence to industry best practices, and compliance with security and performance standards.		
Senior Software Engineer Navigation Team (Automotive Infotainment)	Bosch Global Software Technologies Bengaluru, Karnataka, India	Aug 2016 - Dec 2021
- Developed advanced infotainment features for routing/re-routing using C++ on embedded Linux platforms, boosting navigation accuracy and cutting route recalculation time by 81%. - Conducted Gerrit code review sessions for a team of 8 engineers, using memory-safe C++ design patterns and rigorous quality standards, reducing production defects by 70%. - Collaborated with multiple teams (testing, QA, integration) to deliver high-quality software features on time and within scope, aligning with client and business requirements. - Mentored and trained junior developers through guidance on debugging practices, embedded systems development, and coding standards, improving team capability and speeding up project delivery timelines. - Contributed to the full software development lifecycle (SDLC): requirements gathering, architectural design, implementation, integration, and deployment.		

Research Experience

Research Assistant Brain and Memory Care Team	Healthcare Innovation and Design Lab Santa Clara University, CA, USA	Mar 2024 - Jun 2025
- Developed VR and neurofeedback projects (“Mandala Flow State,” “Fabled Forest”), integrating biofeedback and gesture-based input for apps to boost mindfulness and stress management. . - Presented research on “Fabled Forest” at the MeaningfulXR Conference 2025 (UC Davis) - Won 2nd place in the Child Mind Institute’s 10-Minute Challenge for a GenAI digital intervention (article on the challenge can be accessed here.) - Authored technical documentation; translated EEG signal-processing concepts to non-technical stakeholders.		

Education and Certifications

- AWS Certified Machine Learning - Specialty ([link to certificate](#)) Oct 2025
- AWS Certified AI Practitioner ([link to certificate](#)) Sep 2025
- M.S. Computer Science and Engineering, Santa Clara University, USA. Sep 2023 – Jun 2025
- Nanodegree in Python for Data Science (Udacity) Jul 2022
- B.E. Computer Science and Engineering, PES University, India. Aug 2012 – Sep 2016

Technologies and Languages

- Languages: Python, C, C++, Java, C#, SQL, Dart, HTML/CSS, Node.js
- Technologies: Git, Docker, AWS, GCP, Azure, Linux, Kubernetes, MongoDB, Gerrit, JIRA, Flutter
- AI/ML: TensorFlow, PyTorch, scikit-learn, NumPy, Pandas, HuggingFace, CrewAI
- AR/VR: Unity2D, Unity3D, Meta Quest SDK, Grokit Core
- SWE Practices: Version control, Code review, Debugging, Testing, Agile methods, Microservices, DevOps, CI/CD

Projects

- **Voxure - The Multimodal GenAI Vocal Coach:**
 - Developed cross-platform Flutter application using Computer Vision, LLMs, and RAG to deliver real-time multimodal vocal coaching.
 - Optimized PyTorch models to achieve 90–95% accuracy in posture and voice evaluation.
 - Code can be accessed through [this link](#).
- **Odyssey - The Mobile App for Solo Travelers:**
 - Created the “Connect” module of a cross-platform mobile application enabling users to discover destinations, connect with fellow travelers, share reviews and safety tips.
 - Optimized sorting algorithms and applied design principles for maintainable, scalable and efficient code.
 - Code can be accessed through [this link](#).
- **Abstractive Summarization of News Articles:**
 - Fine-tuned HuggingFace transformer models (BART, Pegasus, Marian) to generate abstractive summaries of long-form news articles, improving ROUGE scores (38–800%) and BLEU scores (97–515%).
 - Researched, implemented, and evaluated advanced neural network optimization techniques and performance metrics to enhance model accuracy, efficiency, and scalability.
- **Secure P2P Storage Component for OpenStack:**
 - Created P2P-based architecture for OpenStack’s storage component, using the Sechord algorithm for handling security issues.
 - File transfer/upload/download times improved by 80%, with quick and reliable detection of suspicious activity.
 - Code can be accessed through [this link](#).

Community and Leadership

- Active participant in coding competitions (Kaggle, Devpost) focused on AI/ML, AR/VR and Data Science.
- Member of “Rewriting the Code”, advancing and collaborating with women in technology.
- Volunteered as a tutor in Computer Science for underprivileged children in Indian schools.
- Contributor/mentee for GirlScript Summer of Code (GSSoC) 2026 for the Open Source & AI Agent tracks.

Interests

- I'm an experienced vocalist trained in Hindustani classical music: performed at private concerts in and around India and the USA.
- Skilled at writing poems, essays, and articles, and served as a school/collegiate magazine editor.
- Other interests include nature photography, painting, drawing and trivia/quizzing.